

Lido Circuit Breaker Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Lido on Circuit Breaker. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated May 1, 2026, was delivered to Lido, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 2 days, the Cyfrin team conducted an audit on the Circuit Breaker code provided by Lido. The findings consist of 2 Informational findings & 3 Gas Optimizations.

Summary

Project Name	Circuit Breaker
Repository	circuit-breaker
Commit	a9e4d1e30faf...
Fix Commit	b4b2fbc921b3...
Audit Timeline	Apr 6th - Apr 7th, 2026
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	0	0	0
Informational	2	1	1
Gas Optimizations	3	2	1
Total	5	3	2

6 Audit Scope

The audit scope was limited to:

```
src/CircuitBreaker.sol
src/Registry.sol
```

7 Deployment Bytecode Verification

Cyfrin was asked to verify that at commit b4b2fbc the locally compiled bytecode for `CircuitBreaker` matched the bytecode deployed to Ethereum mainnet at 0x6019cb557978296ba3c08a7b73225c0975dfb2f7. The following reproducible process was used.

First, prepare the repository:

```
gh repo clone lidofinance/circuit-breaker
cd circuit-breaker
git checkout b4b2fbc921b3191560a3fc62d502d4bb98ad99e1
forge install
forge build
```

Then fetch the deployed bytecode with `eth_getCode` and compare it with the `deployedBytecode` of the local build artifact `out/CircuitBreaker.sol/CircuitBreaker.json`, after stripping the CBOR metadata and ignoring the positions where the local bytecode holds zero bytes for immutable values.

The comparison succeeded. The only differences were at the positions of known immutable values (an address and the pause duration and heartbeat interval bounds). Each immutable was independently confirmed by reading its on-chain getter with `cast call 0x6019cb557978296ba3c08a7b73225c0975dfb2f7 "<getter>"`; the bounds are:

- `MIN_PAUSE_DURATION` = 432000 (5 days)
- `MAX_PAUSE_DURATION` = 5184000 (60 days)
- `MIN_HEARTBEAT_INTERVAL` = 2592000 (30 days)
- `MAX_HEARTBEAT_INTERVAL` = 94608000 (1095 days)

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

`CircuitBreaker` is an emergency pause manager for the Lido protocol: a single contract through which critical Lido contracts can be paused for a globally configured pause duration, after which they resume automatically.

The protocol consists of two contracts:

- `CircuitBreaker` — the main contract, which holds the pause duration and heartbeat interval configuration (within deployment-time min/max bounds), keeps the assignment of a pauser to each pausable contract, and pauses target contracts through their `pauseFor` function.
- `Registry` — a library managing an enumerable pausable-to-pauser registry using a swap-and-pop pattern for O(1) additions and removals.

The pausable contracts this system is expected to manage on Ethereum mainnet include WithdrawalQueue, ValidatorExitBusOracle, TriggerableWithdrawalsGateway, VaultHub, PredepositGuarantee, and CSFeeOracle, all of which implement `pauseFor(uint256)` and `isPaused()` via Lido's `PausableUntil` base contract.

10 Attestation

Cyfrin attests that it performed a security review of the Circuit Breaker code provided by Lido.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-05-01-cyfrin-lido-circuit-breaker-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Lido has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=0b1dc92c84cd304714ccb9bceadf1dc904e05cae30aa34ab5aafa0bfde514d18
F=2026-05-01-cyfrin-lido-circuit-breaker-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.